

Homework 12

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Problem 1. *Prove L_1, L_2, L_3, L_4 are Compact oriented 3 manifolds*

Proof. It suffices to show the conditions for L_1 and then invoking problem 2, it will apply for them all.

1. $L_1(p, q)$ is the image of a surjective map from S^3 a compact space and thus is compact.
2. The action of $\mathbb{Z}_p$ on S^3 has the property that $\forall g \in \mathbb{Z}_p \setminus 0, \forall y \in S^3, \exists U \in \mathcal{T}_{S^3} \ni y \in U$ and $U \cap gU = \emptyset$, this is because if you look at the action of $\mathbb{Z}_p$ on the torii deconstruction of S^3 , you see each point travels around the solid tori (maybe with a twist), and so if you take U to be a small enough open ball, the action of $\forall g \in \mathbb{Z}_p \setminus \{0\}$ will ensure $U \cap gU = \emptyset$. Thus $\varphi : S^3 \rightarrow L_1(p, q)$ is a covering space map. For all $x \in L_1(p, q) \exists U_x \in \mathcal{T}_{L_1(p, q)} \ni x \in U_x$ and $\varphi^{-1}(U_x)$ is a disjoint union of homeomorphic copies of U_x . By surjectivity, $\exists x_0 \in \varphi^{-1}(x_0)$, since S^3 is a 3 manifold $\exists V_{x_0} \in \mathcal{T}_{S^3} \ni x_0 \in V_{x_0} \implies x \in \varphi(V_{x_0})$ where V_{x_0} is homeomorphic to $\mathbb{R}^3$. There exists a homeomorphic copy of U_x, U_{x_0} , inside $\varphi^{-1}(U_x)$ such that $x_0 \in U_{x_0}$. Then $\varphi|_{U_{x_0} \cap V_{x_0}}$ is a homeomorphism and since $L_2(p, q)$ is a quotient map V_{x_0} is open so $U \cap \varphi(V_{x_0})$ is an open set homeomorphic to $\mathbb{R}^3$. Additionally S^3 is 2^{nd} countable and since $L_1(p, q)$ is the image of a surjective map from a second countable space, $L_1(p, q)$ is second countable.

$\forall U \in \mathcal{T}_{S^3}, \varphi^{-1}(\varphi(U)) := \{gx \mid g \in G, x \in U\} = \bigcup_{g \in \mathbb{Z}_p} gU$ is open. Thus $\varphi(U)$ is open. Thus φ is an open map.

$\forall x, y \in S^3 \ni \mathbb{Z}_p x \neq \mathbb{Z}_p y$ (orbits are not equal). Thus $\forall g \in \mathbb{Z}_p, \exists U_g, V_g \in \mathcal{T}_{S^3} \ni gx \in U_g$ and $y \in V_g$ and $U_g \cap V_g = \emptyset$. Then let $U = \bigcap_{h \in \mathbb{Z}_p} h^{-1}U_h$ and $V = \bigcap_{h \in \mathbb{Z}_p} V_h$. Thus $x \in U$ and $y \in V$ and gU is a neighborhood of gx within U_g thus $gU \cap V = \emptyset$ for all g .

Assume the contrary, $\exists x \in \varphi(U) \cap \varphi(V) = \varphi(U \cap V) \implies \exists y \in S^3$ and $g \in G \ni gy \in U \cap V$ but that is a contradiction. Thus $L_1(p, q)$ is Hausdorff. $L_1(p, q)$ is a 3 manifold.

3. Since the action of $\mathbb{Z}_p$ on S^3 is a rotation with a twist, it is orientation preserving. So the quotient space $L_1(p, q)$ must be orientable as it will inherit the orientation of S^3 . For each $x \in S^3$ there is an open set U which doesn't have overlap when acted by $\mathbb{Z}_p$ other than identity. Since S^3 has a consistent global orientation the orientation of each U is the same so when you map it down to $L_1(p, q)$ the orientation must be the same for all of them either way because since $\mathbb{Z}_p$ acts in an orientation preserving way gU has the same orientation as U . So the orientations of each open set of $L_1(p, q)$ is consistent so orientation of $L_1(p, q)$ is consistent globally. Thus $L_1(p, q)$ is orientable.

Thus $L_1(p, q)$ is a compact orientable 3 manifold. Thus by 2, $L_2(p, q), L_3(p, q), L_4(p, q)$ are also compact orientable 3 manifold.

□

Problem 2. Show $L_1(p, q) \cong L_2(p, q) \cong L_3(p, q) \cong L_4(p, q)$

Proof. Let $\omega = e^{2\pi i/p}$.

1. $L_1 \cong L_2$: Let $\varphi : L_2(p, q) \rightarrow L_1(p, q)$ where $[(z, t)] \rightarrow \left[\left(\sqrt{1 - |z|^2} e^{-\frac{\pi i t}{p}}, z \right) \right]$

Let's see this well defined:

(a) $|t| < \sqrt{1 - |z|^2}$: $[(z, t)] = \{(z, t)\}$. Thus φ is well defined for this case.

(b) $t = -\sqrt{1 - |z|^2}$: $[(z, t)] := \{(z, t), (\omega^q z, -t)\}$. Thus

$$\begin{aligned} \varphi([\omega^q z, -t]) &= \left[\left(\sqrt{1 - |z|^2} e^{\frac{\pi i t}{q}}, \omega^q z \right) \right] \\ &= \left[\left(\sqrt{1 - |z|^2} e^{\frac{\pi i t}{p}} \omega^{-1}, z \right) \right] \\ &= \left[\left(\sqrt{1 - |z|^2} e^{-\pi i \frac{t}{p}}, z \right) \right] \\ &= \varphi([(z, t)]) \end{aligned}$$

(c) $t = \sqrt{1 - |z|^2}$: $[(z, t)] := \{(z, t), (\omega^{-q} z, -t)\}$

$$\begin{aligned} \varphi[(\omega^{-q} z, -t)] &= \left[\left(\sqrt{1 - |z|^2} e^{\frac{\pi i t}{p}}, \omega^{-q} z \right) \right] \\ &= \left[\left(\sqrt{1 - |z|^2} e^{\frac{\pi i t}{p}} \omega^{-1}, z \right) \right] \\ &= \left[\left(\sqrt{1 - |z|^2} e^{-\frac{\pi i t}{p}}, z \right) \right] \\ &= \varphi([(z, t)]) \end{aligned}$$

Thus φ is well defined.

Let $\rho : L_1(p, q) \rightarrow L_2(p, q)$ where $[z_1, z_2] \rightarrow [z_2, -\frac{p}{\pi} \arg z_1]$. Thus $(\rho \circ \varphi)(z, t) = \rho[\sqrt{1 - |z|^2} e^{-\frac{i \pi t}{p}}, z] = [z, t]$. Thus φ is injective.

Lets see its surjective. $\forall (z_1, z_2) \in S_3$. Let $z = z_2$ and let $t = -\frac{p}{\pi} \arg(z_1)$. Then, $\varphi([(z, t)]) = [(z_1, z_2)]$. Thus φ is surjective. Since L_2 is the quotient from a compact space B , it is compact. As proven in the previous problem L_1 is Hausdorff. Thus φ is a homeomorphism.

2. $L_1 \cong L_3$: Let $T_1 = \left\{ (z_1, z_2) \mid |z_2|^2 \leq \frac{1}{2}, |z_1|^2 + |z_2|^2 = 1 \right\}$

and $T_2 = \left\{ (z_1, z_2) \mid |z_1|^2 \leq \frac{1}{2}, |z_1|^2 + |z_2|^2 = 1 \right\}$. $S^3 \cong T_1 \cup T_2$. T_1 and T_2 are solid tori. Let m_i, l_i be the meridian and longitude of T_i . $m_1 = z_1 \times S^1$ and $l_1 = S^1 \times z_2$ and $m_2 = S^1 \times z_2$ and $l_2 = z_1 \times S^1$.

In standard S^3 , $m_1 \simeq l_2$ and $l_1 \simeq m_2$. But now lets see what changes by the action of $\mathbb{Z}_p$ on T_1 and T_2 . The classes of orbits of the action, $\frac{1}{p}$ th chunks of the tori where the end faces are

identified. More details are provided in the picture corresponding to this problem. $T_1/\mathbb{Z}_p$ has the following equivalence:

$$(z_1, z_2) \sim (\omega z_1, \omega^q z_2)$$

Similarly $T_2/\mathbb{Z}_p$ has the following equivalences:

$$(z_1, z_2) \sim (\omega^k z_1, \omega z_2)$$

where $k = q^{-1} \in \mathbb{Z}_p^\times$. This causes $T_1/\mathbb{Z}_p$ and $T_2/\mathbb{Z}_p$ to be chunks of T_1 and T_2 where the end faces are identified. Thus $T_1/\mathbb{Z}_p \cong D^2 \times S^1 \cong T_2/\mathbb{Z}_p$. Under the action of $\mathbb{Z}_p$ we will also examine the meridians and longitude to see how they transform under the action

$$\begin{aligned} m_1 &\rightarrow m_1 \\ l_1 &\rightarrow pl_1 + qm_1 \\ m_2 &\rightarrow m_2 \\ l_2 &\rightarrow km_2 + pl_2 \end{aligned}$$

Since $m_2 \sim l_1$ in S^3 , in $L_1(p, q)$, $m_2 \sim pl_1 + qm_1$ and similarly $m_1 \sim km_2 + pl_2$. $L_1(p, q)$ is the disjoint union of two solid tori $T_1/\mathbb{Z}_p$ and $T_2/\mathbb{Z}_p$ with the identification $m_2 \sim pl_1 + qm_1$. This is by definition $L_3(p, q)$.

3. $L_3 \cong L_4$: Let K be the unknot. By definition of Dehn Surgery, you attach $S^3 - K \cong D^2 \times S^1 = T_1$ to a tubular neighborhood of K which is homeomorphic to $D^2 \times S^1 = T_2$ by a map which sends $m_2 \rightarrow qm_1 + pm_2$. Where m_i and l_i are the meridian and the longitude of T_i . This is by definition, the description of L_3 . Thus $L_4 \cong L_3$.

□

Problem 3. Compute the Homology Groups of $L_2(p, q)$.

Proof. The CW structure of $L_2(p, q)$ is as follows, a 0 cell, a 1 cell, a 2 cell, and a 3 cell. Thus the cellular chain complex is as follows:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d_3} \mathbb{Z} \xrightarrow{d_2} \mathbb{Z} \xrightarrow{d_1} \mathbb{Z} \longrightarrow 0$$

Note the map that attaches the 2 cell to the 1 cell is a p sheeted covering of the 1 cell.

$$\begin{aligned} d_1 &= 0 \\ d_2 &= p \end{aligned}$$

So, $H_0(L_2(p, q)) = \mathbb{Z}$, $H_1(L_2(p, q)) = \mathbb{Z}/p\mathbb{Z} = \mathbb{Z}_p$, $H_2(L_2(p, q)) = \ker d_2 / \text{Im } d_3 = 0$. Furthermore $\ker d_2 = 0$ and since $\text{Im } d_3 \subseteq \ker d_2 \implies \text{Im } d_3 = 0$. Thus $d_3 = 0$. Thus $H_3(L_2(p, q)) = \mathbb{Z}$. □

Problem 4. Compute the Fundamental Group of $L_1(p, q)$

Proof. The action of $\mathbb{Z}_p$ on S^3 has the property that $\forall g \in \mathbb{Z}_p \setminus 0, \forall y \in S^3, \exists U \in \mathcal{T}_{S^3} \ni y \in U$ and $U \cap gU = \emptyset$, this is because if you look at the action of $\mathbb{Z}_p$ on the torii deconstruction of S^3 , you see each point travels around the solid tori (maybe with a twist), and so if you take U to be a small enough open ball, the action of $\forall g \in \mathbb{Z}_p \setminus \{0\}$ will ensure $U \cap gU = \emptyset$. Furthermore note $S^3/\mathbb{Z}_p = L_1(p, q)$. Additionally let $p : S^3 \rightarrow L_1(p, q)$ be the quotient map. $\pi_1(S^3) = 0$. By Proposition 1.40, $\mathbb{Z}_p \cong \pi_1(S^3/\mathbb{Z}_p)/\pi_1(S^3) \cong \pi_1(L_1(p, q))$. $\square$

Hatcher Problems

Claim 2.2.18A. For (X, A) a CW pair you have the following:

1. $H_k(X^n, X^{n-1} \cup A^n)$ is free if $k = n$ otherwise is trivial.
2. $H_k(X^n \cup A^{n+1}, A^{n+1}) = 0$ for all $k > n$
3. $X^n \hookrightarrow X$ and $A^n \hookrightarrow A$ induce isomorphisms between $H_n(X^{n+1}, A^{n+1})$ and $H_n(X, A)$
4. $H_n(X^n \cup A^{n+1}, A^{n+1}) \cong H_n(X^n, A^n)$

Proof. 1. $(X^n, X^{n-1} \cup A^n)$ is a CW pair and thus a good pair. Thus $H_k(X^n, X^{n-1} \cup A^n) \simeq H_k(X^n/(X^{n-1} \cup A^n))$. $X^n/(X^{n-1} \cup A^n)$ is a wedge of S^n , where the number of S^n is the number of n -cells which is in X^n but not in A^n . Thus $H_k(X^n/(X^{n-1} \cup A^n)) = \bigoplus H_k(S^n)$ which is $\mathbb{Z}$ when $k = n$ and 0 otherwise. $H_k(X^n, X^{n-1} \cup A^n)$ is free if $k = n$ otherwise is trivial.

2. Let $A = A^{n+1}$ and $B = X^n$, by excision $H_k(X^n \cup A^{n+1}, A^{n+1}) \simeq H_k(X^n, A^n)$. (X^n, A^n) is a CW pair and thus is a good pair. Thus $H_k(X^n, A^n) = H_k(X^n/A^n)$. X^n/A^n is a n dimensional CW complex. By Lemma 2.34b, $H_k(X^n/A^n) = 0$.

3. X^{n+1}/A^{n+1} is a $n+1$ dimensional CW Complex, where its $n+1$ skeleton is the $n+1$ skeleton of X^{n+1} where A^{n+1} skeleton have been identified to a point. (X/A) is also a CW complex, where A in X has been identified to a point. But note $(X/A)^{n+1}$ or the $n+1$ skeleton of X/A is the $n+1$ skeleton of X , ie X^{n+1} , which is not in A and A is reduced to a point. Any cell in the X^{n+1} is either in A or not in A . If its in A , its in A^{n+1} . And if its not in A its not in A^{n+1} . $(X/A)^{n+1}$ is the n skeleton of X^{n+1} where A has been reduced to a point, it is the same thing as X^{n+1} where A^{n+1} is reduced to a point, this is by definition the same thing as X^{n+1}/A^{n+1} . Furthermore (X, A) and (X^{n+1}, A^{n+1}) are good pairs, and by 2.34(c), $H_n(X/A) \cong H_n((X/A)^{n+1})$. Thus we have

$$H_n(X^{n+1}, A^{n+1}) \cong H_n(X^{n+1}/A^{n+1}) \cong H_n((X/A)^{n+1}) \cong H_n(X/A) \cong H_n(X, A)$$

4. Let $A = A^{n+1}$ and $B = X^n$. By excision, $H_n(X^n \cup A^{n+1}, A^{n+1}) \cong H_n(B, A \cap B) = H_n(X^n, A^n)$. $\square$

Problem 2.2.18. For (X, A) , a CW Pair, show there is a relative cellular chain complex formed by groups isomorphic to $H_i(X^i, X^{i-1} \cup A^i)$

Proof. Use Lemma 2.2.18A, to create this commutative diagram using the long exact sequences of triples $(X^{n+1}, X^n \cup A^{n+1}, A^{n+1})$, $(X^n, X^{n-1} \cup A^n, A^n)$, $(X^{n-1}, X^{n-2} \cup A^{n-1}, A^{n-1})$. Let d_n be the maps which make the diagram commute.

$$\begin{array}{ccccccc}
& & H_n(X^{n+1}, X^n \cup A^{n+1}) = 0 & & & & \\
& & \uparrow & & & & \\
& & H_n(X^{n+1}, A^{n+1}) & & H_n(X^{n-1} \cup A^n, A^n) = 0 & & \\
& & \uparrow e & & \downarrow & & \\
& & H_n(X^n \cup A^{n+1}, A^{n+1}) & \cong & H_n(X^n, A^n) & & H_{n-2}(X^{n-2} \cup A^{n-1}, A^{n-1}) \\
& & \uparrow c & & \downarrow b & & \uparrow \\
\cdots \longrightarrow & H_{n+1}(X^{n+1}, X^n \cup A^{n+1}) & \xrightarrow{d_{n+1}} & H_n(X^n, X^{n-1} \cup A^n) & \xrightarrow{d_n} & H_{n-1}(X^{n-1}, X^{n-2} \cup A^{n-1}) & \longrightarrow \cdots \\
& & & \downarrow a & & \uparrow & \\
& & & H_{n-1}(X^{n-1} \cup A^n, A^n) & \cong & H_{n-1}(X^{n-1}, A^{n-1}) & \\
& & & & & \uparrow & \\
& & & & & 0 &
\end{array}$$

By injectivity and exactness, $\ker d_n = \ker a = \text{Im} b \cong H_n(X^n, A^n) \cong H_n(X^n \cup A^{n+1}, A^{n+1})$.

Again by injectivity and exactness, $\text{Im} d_{n+1} \cong \text{Im}(c) = \ker e$. Thus the homotopy group of the cellular chain formed by $H_n(X^n, X^{n-1} \cup A^n)$ is $H_n(X^n \cup A^{n+1}, A^{n+1}) / \ker e \cong H_n(X^{n+1}, A^{n+1})$ by isomorphism theorem of groups since e is surjective. By the Lemma 2.2.18A, $H_n(X^{n+1}, A^{n+1}) = H_n(X, A)$. Thus the homotopy group of the cellular chain formed by $H_n(X^n, X^{n-1} \cup A^n)$ is $H_n(X, A)$. □

Problem 2.2.20. Suppose X and Y are finite CW complexes.

$$\chi(X \times Y) = \chi(X)\chi(Y)$$

Proof. For a arbitrary CW complex A , let c_i^A be the number i th cells in A .

By Theorem A.6, $X \times Y$ has the CW structure where the characteristic maps of the structure are $\varphi_\alpha \times \psi_\beta$ where φ_α are the characteristic maps of X and ψ_β are the characteristic maps of Y . Thus

$$c_k^{X \times Y} = \sum_{i+j=k} c_i^X c_j^Y$$

Thus,

$$\begin{aligned}
\chi(X \times Y) &= \sum_k (-1)^k c_k^{X \times Y} \\
&= \sum_k (-1)^k \sum_{i+j=k} c_i^X c_j^Y \\
&= \sum_k \sum_{i+j=k} (-1)^{i+j} c_i^X c_j^Y \\
&= \left(\sum_i (-1)^i c_i^X \right) \left(\sum_j (-1)^j c_j^Y \right) \\
&= \chi(X) \chi(Y)
\end{aligned}$$

□

Problem 2.2.21. A finite CW complex X , is a union of subcomplexes A and B , show that $\chi(X) = \chi(A) + \chi(B) - \chi(A \cap B)$

Proof. Any cell must be at least A or B . By law of inclusion exclusion,

$$c_k^X = c_k^A + c_k^B - c_k^{A \cap B}$$

Thus,

$$\begin{aligned}
\chi(X) &= \sum_k (-1)^k c_k^{X \times Y} \\
&= \sum_k (-1)^k (c_k^A + c_k^B - c_k^{A \cap B}) \\
&= \left(\sum_k (-1)^k c_k^A \right) + \left(\sum_k (-1)^k c_k^B \right) - \left(\sum_k (-1)^k c_k^{A \cap B} \right) \\
&= \chi(A) + \chi(B) - \chi(A \cap B)
\end{aligned}$$

□

Claim 2.2.22. For X a finite CW complex and $p : Y \rightarrow X$ an n -sheeted covering, show that $\chi(Y) = n\chi(X)$

Proof. Let $\varphi_\alpha : e_\alpha^n \rightarrow X$ be a characteristic map. Since $\pi_1(e_\alpha^n, *) = 0$, there is a lift $\varphi'_\alpha : e_\alpha^n \rightarrow Y$ and thus each $p^{-1}(e_\alpha^n)$ is homeomorphic to e_α^n . Since $p : Y \rightarrow X$ is a n -sheeted covering $p|_{e_\alpha^n}$ is a n -sheeted covering of e_α^n . Thus each cell has n copies. Thus $c_k^Y = c_k^X \times n$.

Thus $\chi(Y) = \sum_k (-1)^k c_k^Y = n \sum_k (-1)^k c_k^X = n\chi(X)$.

□

Claim 2.2.24. Show that if the closed orientable surface M_g of genus g is a covering space of M_h , $g = n(h - 1) + 1$ where n is the number of sheets in the covering.

Proof. $\forall g \in \mathbb{N} \cup \{0\}$, $\chi(M_g) = 2 - 2g = 2(1 - g)$ as previously proven.

Suppose M_g for $g \in \mathbb{N} \cup \{0\}$ is a n sheeted covering of M_h . Then by 2.2.22,

$$2 - 2g = \chi(M_g) = n\chi(M_h) = 2n(1 - h)$$

$$\text{Thus } g = \frac{2n(h-1)+2}{2} = n(h - 1) + 1$$

□

Problem 2.2.28a. Use Mayer Vietoris to compute the homology groups of space of $S^1 \times S^1 \sqcup M$ where boundary circle of M is identified to $S^1 \times \{x_0\}$

Proof. Let A be the torus plus a tiny portion of the mobius strip which deformation retracts to the torus and let B be the mobius strip plus a tiny portion of the torus which deformation retracts to the mobius strip.

Note:

$$\tilde{H}_n(S^1 \times S^1) = \begin{cases} \mathbb{Z}^2 & \text{if } n = 1 \\ \mathbb{Z} & \text{if } n = 2 \\ 0 & \text{otherwise} \end{cases}$$

Additionally Note that M deformation retracts to center circle.

Furthermore note there are no 3 cells. So we have the following Mayer Vietoris Sequence:

$$0 \longrightarrow \mathbb{Z} \oplus 0 \hookrightarrow \tilde{H}_2(X) \xrightarrow{j} \mathbb{Z} \xrightarrow{i} \mathbb{Z}^2 \oplus \mathbb{Z} \longrightarrow \tilde{H}_1(X) \longrightarrow 0$$

Note the generator of $A \cap B$ goes around once $S^1 \times \{z_0\}$ in A and 2 times around B .

$$\text{Thus } i = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}. \text{ Thus } H_1(X) \cong \tilde{H}_1(X) = \mathbb{Z}^2 \oplus \mathbb{Z} / \text{span} \left(\begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} \right) = \mathbb{Z}^2 \oplus \mathbb{Z}_2$$

i is injective, thus by exactness $\text{Im } j = \ker i = 0$. Thus we have the exact sequence:

$$0 \longrightarrow \mathbb{Z} \oplus 0 \hookrightarrow \tilde{H}_2(X) \xrightarrow{0} 0$$

Thus $\mathbb{Z} \cong H_2(X) \cong \tilde{H}_2(X)$. Furthermore X is connected, so $H_0(X) = \mathbb{Z}$. Furthermore for all $n > 2$, $H_n(X) = 0$. □

Problem 2.2.28a. Use Mayer Vietoris to compute the homology groups of space of $\mathbb{R}P^2$ where boundary circle of M is identified to $\mathbb{R}P^1$

Proof. Let A be the $\mathbb{R}P^2$ plus a tiny portion of the mobius strip which deformation retracts to the torus and let B be the mobius strip plus a tiny portion of the torus which deformation retracts to the $\mathbb{R}P^2$.

There are no 3 cells. So we have the following Mayer Vietoris Sequence:

$$0 \longrightarrow \tilde{H}_2(X) \hookrightarrow \mathbb{Z} \xrightarrow{i} \mathbb{Z}_2 \oplus \mathbb{Z} \longrightarrow \tilde{H}_1(X) \longrightarrow 0$$

Again generator of $A \cap B$, goes once around A and 2 times around B . Thus $i = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. i is injective. By exactness, $H_2(X) = \tilde{H}_2(X) = \ker i = 0$. Furthermore, $H_1(X) = \tilde{H}_1(X) = \frac{\mathbb{Z}_2 \oplus \mathbb{Z}}{\text{span}\left(\begin{bmatrix} -1 \\ 2 \end{bmatrix}\right)} = \mathbb{Z}_2 \oplus \mathbb{Z}_2$.

X is connected so $H_0(X) = \mathbb{Z}$ and for all $n > 2$ $H_n(X) = 0$.

□

Problem 2.2.29a. X is 2 solid genus g solids with boundary genus g surface identified by identity. Let R be the 2 solid genus g solid

Proof. Let A be a open set which deformation retracts to one copy of R and let B be the open set which deformation retracts to the other copy of R . Note R deformation retracts to chain of S^1 's which is homotopy equivalent to wedge of g S^1 's. Thus $R \simeq \bigvee_{k=1}^g S^1$.

Note X is connected so $H_0(X) = \mathbb{Z}$. $A \cap B \simeq M_g$.

By Meyer Vietoris:

$$0 \longrightarrow \oplus_{i=1}^2 \tilde{H}_3(R) \cong 0 \longrightarrow \tilde{H}_3(X) \hookrightarrow \mathbb{Z} \longrightarrow \oplus_{i=1}^2 \tilde{H}_2(R) \cong 0$$

$$\oplus_{i=1}^2 \tilde{H}_2(R) \cong 0 \longrightarrow \tilde{H}_2(X) \xhookrightarrow{j} \mathbb{Z}^{2g} \xrightarrow{i} \oplus_{i=1}^2 \tilde{H}_1(R) \cong \mathbb{Z}^{2g} \xrightarrow{k} \tilde{H}_1(X) \longrightarrow 0$$

The generator of $H_2(M_g)$ maps to 1 and -1 in A and B respectively so i is.

$$i = \begin{bmatrix} 1 & & \dots & 0 & \dots \\ & 1 & \dots & 0 & \dots \\ \vdots & \vdots & \ddots & \vdots & \\ -1 & & \dots & 0 & \dots \\ & -1 & \dots & 0 & \dots \\ \vdots & \vdots & \ddots & \vdots & \end{bmatrix}$$

Let $a_1, \dots, a_g$ be the generator for $\tilde{H}_1(A)$. Let $b_1, \dots, b_g$ be the generator for $\tilde{H}_1(B)$.

Let $m_1, \dots, m_{2g}$ be the generator for $\tilde{H}_1(M_g)$.

i sends $m_i \rightarrow a_i - b_i$ for $i \in [1, \dots, g]$ and $m_i \rightarrow 0$ otherwise.

Thus $H_1(X) = \tilde{H}_1(X) = \frac{\mathbb{Z}^{2g}}{\text{span}(a_1 - b_1, \dots, a_g - b_g)} \cong \mathbb{Z}^g$.

$H_2(X) = \tilde{H}_2(X) = \ker i = \text{span}(m_{g+1}, \dots, m_{2g}) = \mathbb{Z}^g$.

Finally $H_3(X) = \tilde{H}_3(X) = \mathbb{Z}$. Since X is connected $H_0(X) = \mathbb{Z}$. $\forall n > 3$, $H_n(X) = 0$.

□

Problem 2.2.29b. Find $H_k(R, M_g)$

Proof. Note R deformation retracts to chain of S^1 's which is homotopy equivalent to wedge of g S^1 's. Thus $R \simeq \bigvee_{k=1}^g S^1$. We have the following exact sequence but note i_* is the induced map from the inclusion of $i: M_g \hookrightarrow R$. By Lemma 2.34(c), i is surjective.

$$0 \longrightarrow H_2(R, M_g) \hookrightarrow \mathbb{Z}^{2g} \xrightarrow{i_*} \mathbb{Z}^g \xrightarrow{j} H_1(R, M_g) \longrightarrow 0$$

Since $\ker j = \operatorname{Im} i_* = \mathbb{Z}_g$ and j is surjective, $H_1(R, M_g) = 0$. Additionally i_* is the map which identifies the meridian and longitude of each "torus", section, of the M_g surface. Or notationally,

$$i_* = \begin{bmatrix} 1 & -1 & 0 & \dots \\ 0 & 0 & 1 & -1 & \dots \\ \vdots & \vdots & \ddots & \ddots \end{bmatrix}$$

Thus kernel of i_* is the span of all the longitudes. So $H_2(R, M_g) \cong \ker i_* \cong \mathbb{Z}^g$. Finally $H_m(R, M_g) = 0$ for all $m \neq 2$.

□